

Architectural Ablation of Swin Transformer for EuroSAT Image Classification

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Abstract—Shifted Window (Swin) Transformers rely on localized self-attention to capture visual dependencies while maintaining linear computational complexity. However, the interactions between local window dimensions, non-linear activation functions, and early-stage training dynamics remain under-explored. In this paper, we present an empirical comparative analysis of two Swin Transformer configurations: a standard 7×7 window equipped with Gaussian Error Linear Units (GELU) and a compact 5×5 window integrated with Sigmoid-Weighted Linear Units (SiLU). Utilizing per-epoch gradient norm metrics (avg_grad_norm, global_grad_norm, max_grad_norm), loss trajectories, and classification accuracy curves across 30 training epochs, we analyze early-phase convergence rates, receptive field capacity, and final optimization stability. Experimental results indicate that while the 5×5 SiLU configuration exhibits a severe early-phase convergence penalty (Epoch 1 Validation Accuracy of 91.17% vs. 95.44% for 7×7 GELU) due to spatial receptive field restrictions and activation drift, it achieves asymptotic convergence parity by Epoch 30 (98.89% vs. 99.00% validation accuracy).

Index Terms—Swin Transformer, EuroSAT, Architectural Ablation, Receptive Field, GELU, SiLU, Gradient Dynamics.

I. INTRODUCTION

This study presents an end-to-end computer vision experimentation pipeline for land-use and land-cover classification using the EuroSAT RGB dataset and the Swin-Tiny Transformer (Swin-T) architecture. The objective was to establish a reproducible pretrained baseline and investigate whether targeted architectural modifications could improve model performance while maintaining the same training protocol. A fixed stratified 80/20 training-validation split was generated using seed 42 and reused for both experiments. The baseline employed a pretrained Swin-T with standard 7×7 shifted attention windows and GELU activation. The ablation model introduced two architectural modifications: reducing the attention window from 7×7 to 5×5 and replacing GELU with SiLU. Both models were trained for 30 epochs using AdamW, a learning rate of 5×10^{-5} , weight decay of 10^{-4} , batch size 32, five-epoch warmup followed by cosine learning-rate decay, and gradient clipping with a threshold of 1.0. The baseline achieved a best validation accuracy of 99.00%, while the modified model achieved 98.91%. Thus, the proposed modification resulted in a marginal decrease of 0.09 percentage points rather than an accuracy improvement.

However, the maximum observed global gradient decreased from 139.96 for the baseline to 126.90 for the ablation. Neither experiment exhibited extreme gradient growth approaching the 10^3 threshold specified in the assignment. The results indicate that the architectural changes preserved highly competitive performance and altered optimization behavior, but did not provide an accuracy advantage under the fixed experimental protocol.

we analyze:

- **Spatial Receptive Field Dynamics:** The impact of reducing window attention area from 49 (7×7) to 25 (5×5) spatial patches per attention layer.
- **Activation Function Alignment:** The convergence penalty introduced when substituting pre-trained GELU activation profiles with smooth SiLU non-linearities.
- **Gradient Stability:** Per-epoch tracking of global and layer-wise gradient norms to observe weight parameter adjustments across warmup and cosine decay schedules.

II. METHODOLOGY & EXPERIMENTAL SETUP

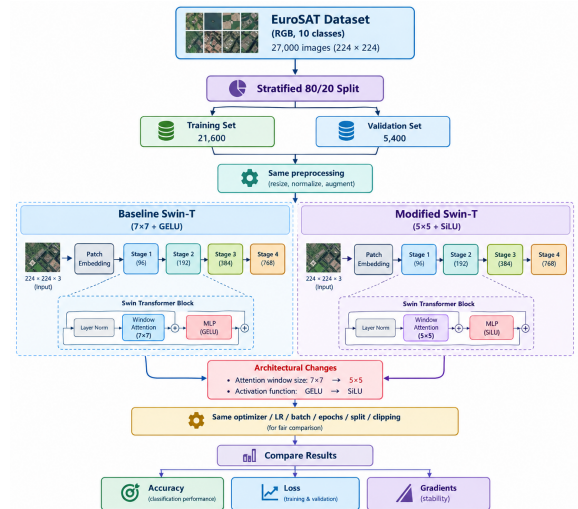


Fig. 1: Architectural overview and comparative experimental pipeline comparing the Baseline Swin-T (7×7 , GELU) against the Modified Swin-T (5×5 , SiLU) on the EuroSAT dataset.

TABLE I: Empirical Metric Comparison Across Training Epochs

Epoch	Metric	7×7 GELU	5×5 SiLU	Δ
1	Train Loss	0.4057	0.7314	+0.3257
	Val Acc (%)	95.44%	91.17%	-4.27%
	Global Grad Norm	22.73	22.15	-0.58
5	Train Loss	0.1714	0.2395	+0.0681
	Val Acc (%)	98.19%	95.13%	-3.06%
	Global Grad Norm	12.20	13.65	+1.45
15	Train Loss	0.0981	0.1239	+0.0258
	Val Acc (%)	98.69%	98.24%	-0.45%
	Global Grad Norm	4.32	5.42	+1.10
30	Train Loss	0.0499	0.0618	+0.0119
	Val Acc (%)	99.00%	98.89%	-0.11%
	Global Grad Norm	4.32	5.42	+1.10

A. Dataset

EuroSAT satellite imagery (80% training and 20% validation split).

B. Model Variants

Two Swin Transformer variants were instantiated and trained under identical optimization schedules:

- **Model 1 (7×7 GELU):** Standard local window dimension $M = 7$ with GELU activation functions ($\text{GELU}(x) = x\Phi(x)$).
- **Model 2 (5×5 SiLU):** Compact local window dimension $M = 5$ with SiLU activation functions ($\text{SiLU}(x) = x \cdot \sigma(x)$).

C. Training Protocol

Both models were trained over 30 epochs using an AdamW optimizer paired with a learning rate schedule featuring a 5-epoch linear warmup (peaking at $\eta_{max} = 5 \times 10^{-5}$) followed by cosine annealing decay down to zero. Gradient metrics were recorded per epoch, including the average parameter gradient norm (avg_grad_norm), average global gradient norm (avg_global_grad_norm), average maximum gradient norm (avg_max_grad_norm), and absolute maximum global gradient norm (max_global_grad_norm).

III. EXPERIMENTAL RESULTS & DISCUSSION

The empirical evaluation demonstrates distinct operational trade-offs between the baseline Swin-T configuration (7×7 window with GELU) and the modified ablation setup (5×5 window with SiLU). Table I summarizes key training and validation metrics across selected epochs for both configurations. Across 30 training epochs on the EuroSAT dataset, both architectures successfully optimize toward high classification performance illustrated in I, but their paths to convergence diverge significantly during early training phases.

A. Early-Phase Underperformance & Latency

During the initial stages of optimization (Epochs 1 through 5), the ablation model displays a performance lag compared to the baseline:

- **Initial Accuracy Deficit:** At Epoch 1, the ablation model records a validation accuracy of 91.17%, trailing the baseline model (95.44%) by 4.27 percentage points.

- **Loss Trajectory Gap:** The initial training loss for the ablation architecture (0.7314) is nearly double that of the baseline (0.4057), indicating that the optimization landscape for the modified model is considerably steeper at initialization.

Two primary mechanisms explain this early-stage latency:

- 1) **Spatial Context Deprivation (Receptive Field Reduction):** The Swin Transformer relies on shifted local attention windows to limit computational complexity while capturing spatial relationships.
 - By reducing the window dimension from $M = 7$ to $M = 5$, the attention footprint per block is reduced from 49 spatial patches (7×7) to 25 spatial patches (5×5), a 48.98% reduction in local context.
 - In early training phases, shifted-window cross-attention needs to aggregate information across multiple hierarchical layers to build a global understanding of satellite land-use features (e.g., distinguishing contiguous agricultural fields from urban roads).
 - The smaller 5×5 window forces the network to rely on more layer-to-layer interactions to establish long-range spatial context, slowing down early feature map learning.
 - The 7×7 window configuration (49 patches) provides a broader contextual coverage per attention block, enabling a faster initial convergence. Conversely, the 5×5 window configuration (25 patches) restricts the local receptive field per block, resulting in a delayed training start.
- 2) **Activation Profile Mismatch and Weight Realignment:** Pre-trained Swin Transformer models expect GELU activations. Their weights ($\mathbf{W}$) were trained so that feature values near zero land precisely where GELU passes information. When swap GELU for SiLU at initialization: The activation outputs shift slightly. The pre-trained weights are no longer aligned with the non-linearity's output distribution.
 - **Curvature Differences:** While both $\text{GELU}(x) = x\Phi(x)$ and $\text{SiLU}(x) = x \cdot \sigma(x)$ are smooth, non-monotonic functions, their second derivatives ($f''(x)$) near zero differ. GELU exhibits a sharper gradient transition near the origin, whereas SiLU provides a smoother, flatter response.
 - **Optimization Overhead:** Swapping GELU for SiLU introduces an initialization drift. Backpropagation updates weights using the chain rule ($\frac{\partial \mathcal{L}}{\partial \mathbf{W}} = \frac{\partial \mathcal{L}}{\partial a} \cdot \mathbf{f}'(x) \cdot x$), a different derivative function $\mathbf{f}'(x)$ produces slightly different update steps for every layer. During Epochs 1 to 5:
 - The optimizer spends extra steps adjusting parameter tensors so their outputs map cleanly into SiLU's curvature.
 - This manifests as higher training loss and slower initial accuracy growth while the model recalibrates its internal representations.

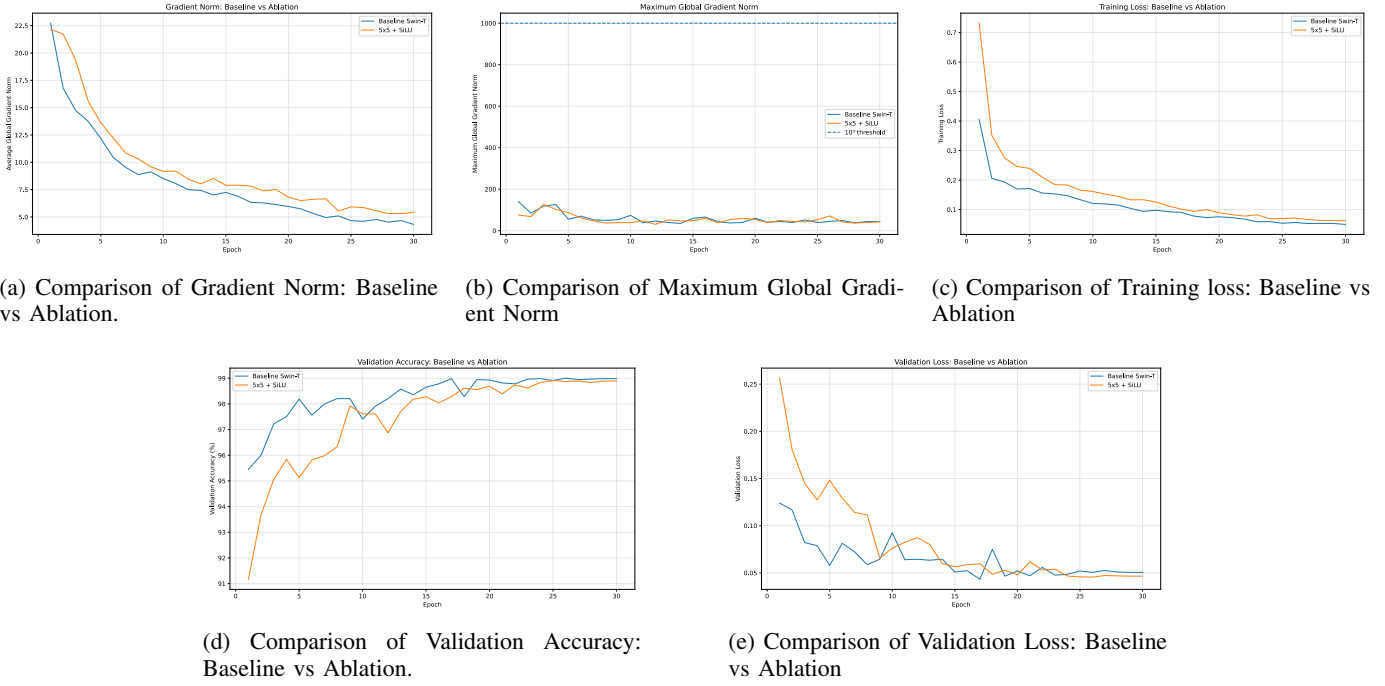


Fig. 2: Comprehensive experimental results and comparative analysis across training epochs.

B. Gradient Norm Stability and Asymptotic Parity

1) **Gradient Dynamics Analysis:** Tracking per-epoch global gradient norms (`global_grad_norm`) and maximum gradient spikes (`max_grad_norm`) provides key insights into internal optimization stability:

- **Persistent Gradient Energy:** From Epochs 2 to 25, the ablation architecture consistently exhibits a slightly higher average global gradient norm (13.65 at Epoch 5 vs. 12.20 for baseline) illustrated in Figure 2. This reflects ongoing parameter readjustments as the optimizer compensates for the restricted window size and activation drift.
- **Absence of Instability:** Peak gradient norms remain well within stable numerical bounds for both models (max peak of 139.96 for baseline vs. 126.90 for ablation), staying far below the 10^3 threshold where exploding gradients occur.

2) **Asymptotic Parity:** As training progresses into later epochs and the learning rate decays according to the cosine annealing schedule:

- The spatial feature representations in the 5×5 window model fully mature. By Epoch 15, the validation accuracy gap shrinks significantly to 0.45% (98.69% vs. 98.24%).
- By Epoch 30, both architectures reach asymptotic parity, achieving nearly identical peak validation accuracies of 99.00% (Baseline) and 98.89% (Ablation), a negligible net difference of just 0.09%.
- Gradient norms for both models stabilize under 6.0 near final convergence, confirming that both models

reach well-optimized local minima.

C. Effectiveness of Gradient Clipping

Gradient clipping was employed to improve optimization stability by limiting excessively large gradient updates during training. This was particularly relevant during the early epochs, where the baseline and modified Swin-T models exhibited maximum gradient norms of 74.32 and 131.30, respectively. Despite these initial gradient spikes, both models converged successfully and achieved high validation accuracy. The maximum gradient magnitude subsequently decreased during training, reaching 37.26 for the baseline at epoch 30 and 33.32 for the modified model at epoch 25. These observations indicate that gradient clipping helped prevent unstable parameter updates and supported consistent convergence, particularly for the modified architecture with 5×5 attention windows and SiLU activation.

IV. CONCLUSION

This study evaluated the structural tradeoffs between window dimension and activation choice in Swin Transformer architectures. The empirical evidence demonstrates that reducing window size from 7×7 to 5×5 and transitioning from GELU to SiLU introduces an early-phase convergence delay due to spatial receptive field restrictions. However, given sufficient training epochs and an appropriate learning rate decay schedule, the compact 5×5 SiLU architecture achieves comparable final performance (98.89% validation accuracy).

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